

Amirhossein Sourì

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Education

Sharif University of Technology <i>B.Sc. in Computer Science and Engineering</i> – GPA: 18.99/20 – Ranked 14th among 145,000+ participants in Iran's National University Entrance Examination.	2022 – Expected June 2027 <i>Tehran, Iran</i>
Shahid Beheshti High School (NODET) <i>Diploma in Mathematics and Physics</i> Affiliated with the National Organization for Development of Exceptional Talents	2019 – 2022 <i>Tehran, Iran</i>

Research Interests

- Machine and deep learning
- Computer vision and multimodal learning
- Language and vision-language models
- ML Systems
- Generative models
- Trustworthy ML
- Reinforcement learning and multi-agent systems
- Robotics

Research Experience

RIML Lab — Visual Reasoning in Vision-Language Models <i>Research Assistant (PI: Dr. Mohammad Hossein Rohban), Sharif University of Technology</i> – Developed the project's initial question-conditioned visual-grounding study using CLIP patch/text features, heatmap-to-box conversion, and downstream evaluation with Qwen2.5-VL-3B-Instruct. – Investigating training and evaluation methods for evidence-sensitive reasoning in vision-language models.	July 2026 – Present
RIML Lab — Diffusion Model Unlearning <i>Research Assistant (PI: Dr. Mohammad Hossein Rohban), Sharif University of Technology</i> – Investigated noise-robust concept unlearning for text-to-image diffusion models under adversarial and distributional variation. – Contributed to adaptive noise sampling and experimental evaluation; this work resulted in a manuscript currently under revision.	Dec 2025 – Mar 2026

Manuscript

Weeding Out Bad Seeds: Noise-Robust Unlearning for Text-to-Image Diffusion Models <i>Co-author; under review at ICLR 2027.</i>	2026
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Teaching Experience

Sharif University of Technology <i>Teaching Assistant</i> – Artificial Intelligence — Dr. Mahdiah Soleymani – Artificial Intelligence — Mr. Samiei and Mr. Fereydooni – Machine Learning — Dr. Abolfazl Motahari – Probability and Statistics — Dr. Amir Najafi – Advanced Programming (Java) — Dr. MohammadAmin Fazli – Fundamentals of Programming (C) — Dr. MohammadAmin Fazli	Oct 2023 – Present <i>Tehran, Iran</i> <i>Fall 2025</i> <i>Spring 2025</i> <i>Spring 2025</i> <i>Spring 2024</i> <i>Spring 2024, Spring 2026</i> <i>Fall 2023, Fall 2024, Fall 2025</i>
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Industry Experience

Hamravesb <i>Software Engineer</i> – Developed and deployed a marketplace service for one-click cloud applications, spanning backend APIs, frontend interfaces, and Kubernetes integration.	Oct 2024 – Sept 2025 <i>Tehran, Iran</i>
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Selected Projects

- Question-Conditioned Visual Grounding | [GitHub](#) July 2026
 - Trained an attention-based model on CLIP patch and question embeddings to localize answer-relevant image regions.
 - Converted predicted heatmaps into bounding boxes and evaluated their effect on downstream VQA with Qwen2.5-VL-3B-Instruct.
- Deep Learning Practical Assignments | [GitHub](#) Spring 2026
 - Completed 15 implementation-focused notebooks spanning neural networks and optimizers from scratch, CNNs/RNNs, LoRA/QLoRA adaptation, RAG, and language model inference.
 - Implemented and evaluated VAE, DDPM, VQ-VAE-2, SimCLR, CLIP analysis, and latent-diffusion image-editing pipelines using PyTorch.
- Modern Information Retrieval Project | [GitHub](#) Spring 2026
 - Built a Goodreads search engine with preprocessing, near-duplicate detection, inverted/tiered indexes, BM25 and language model ranking, and MAP/NDCG/MRR evaluation.
 - Developed a hybrid multimodal product-search pipeline combining dense multimodal embeddings, SPLADE++ sparse retrieval, cross-encoder reranking, and structured LLM responses.
- Machine Unlearning and Robustness Exercises | [GitHub](#) Dec 2025
 - Implemented class unlearning in a conditional VAE using Fisher information and evaluated recovery attacks based on label manipulation, latent interpolation, and condition optimization.
- Artificial Intelligence Practical Assignments | [GitHub](#) Spring 2024
 - Implemented classical search, constraint satisfaction, minimax with alpha-beta pruning, Bayesian-network inference, HMMs, Q-learning, SARSA, and value iteration.
- Machine Learning Models Collection | [GitHub](#) Sept 2023
 - Implemented, explained, and compared core supervised and unsupervised methods—including regression, SVMs, trees, ensembles, dimensionality reduction, clustering, and Gaussian mixtures—with scikit-learn baselines.

Technical Skills

Programming	Python, C/C++ , Java, TypeScript/JavaScript, SQL, R
Machine Learning	PyTorch, Transformers, Diffusers, PEFT, TRL, Accelerate, scikit-learn, SentenceTransformers, TensorFlow
Data & Vision	NumPy, pandas, Hugging Face Datasets, FAISS, OpenCV, Matplotlib, Jupyter
Systems & Software	Linux, Git, Docker, Kubernetes, vLLM, Django/DRF, React, PostgreSQL
Documentation	LaTeX, Markdown

Selected Coursework

Deep Learning	Dr. Mahdieh Soleymani	19.7/20
Machine Learning	Dr. Ali Sharifi Zarchi	20.0/20
Artificial Intelligence	Dr. Mohammad Hossein Rohban	19.3/20
Modern Information Retrieval	Dr. Mahdieh Soleymani	20.0/20
Computer Simulation	Dr. Bardia Safaei	In progress
Linear Algebra	Dr. Maryam Ramezani	20.0/20
Probability and Statistics	Dr. Amir Najafi	19.2/20

Languages

Persian: Native proficiency English: Professional working proficiency

Selected Academic Service & Leadership

Public Relations Officer, CE Students’ Scientific Chapter	July 2025 – Present
Senior Editor, Byte Publication	May 2025 – Present
Vice President & Technical Lead, Hardwar Event	Feb 2025 – May 2025